Exam Papers Practice

OCR Further Mathematics
Pure Core - Series
Question Paper
AS and A Level
Further Mathematics A - H235, H245

The latest topic questions for the year 2024 / 2025+ exams.
To be used by all students preparing for OCR AS and A Level Further Mathematics.
Students of other boards may also find this useful.

1

Using an algebraic method, determine the least value of n for which $\sum_{r=1}^{n} \frac{1}{(2r-1)(2r+1)} \geqslant 0.49$. [8]

2 In this question you must show detailed reasoning.

Determine the value of $\sum_{r=1}^{100} (2r+3)^2$. [3]

3(a) By using partial fractions show that $\sum_{r=1}^{n} \frac{1}{r^2+3r+2} = \frac{1}{2} - \frac{1}{n+2}$. [5]

(b) Hence determine the value of $\sum_{r=1}^{\infty} \frac{1}{r^2+3r+2}$ [2]

4(a) In this question you must show detailed reasoning.

Use partial fractions to show that $\sum_{r=5}^{n} \frac{3}{r^2+r-2} = \frac{37}{60} - \frac{1}{n} - \frac{1}{n+1} - \frac{1}{n+2}$. [5]

(b)

Write down the value of $\lim_{n\to\infty}\left(\sum_{r=5}^{n} \frac{3}{r^2+r-2}\right)$. [1]

5 Prove by induction that the sum of the cubes of three consecutive positive integers is divisible by 9. [5]

6 Find an expression for $1 \times 2^2 + 2 \times 3^2 + 3 \times 4^2 + \ldots + n(n+1)^2$ in terms of n. Give your answer in fully factorised form. [3]

7

Using the formulae for $\sum_{r=1}^{n} r$ and $\sum_{r=1}^{n} r^2$, show that $\sum_{r=1}^{10} r(3r-2) = 1045$. [3]

8

(a) Given that $u = \tanh x$, use the definition of $\tanh x$ in terms of exponentials to show that $x = \frac{1}{2}\ln\left(\frac{1+u}{1-u}\right)$. [4]

(b) Solve the equation $4\tanh^2 x + \tanh x - 3 = 0$, giving the solution in the form alnb where a and b are rational numbers to be determined. [4]

(c) Explain why the equation in part (b) has only one root. [1]

9

(a) Determine an expression for $\sum_{r=1}^{n} \frac{1}{r(r+1)(r+2)}$ giving your answer in the form $\frac{1}{4} - \frac{1}{2}f(n)$. [6]

(b) Find the value of $\sum_{r=1}^{\infty} \frac{1}{r(r+1)(r+2)}$. [1]

10

C is the locus of numbers, z, for which $\mathrm{Im}\left(\frac{z+7\mathrm{i}}{z-24}\right)=\frac{1}{4}$.

By writing $z = x+\mathrm{i}y$ give a complete description of the shape of C on an Argand diagram. [7]

11

(a) Find as a single algebraic fraction an expression for $\sum_{r=1}^{n} \frac{1}{(2r-1)(2r+1)}$. [4]

(b) Determine the value of $\sum_{r=1}^{\infty} \frac{1}{(2r-1)(2r+1)}$. [1]

12

(a) By considering $\sum_{r=1}^{n}\left((r+1)^5-r^5\right)$ show that $\sum_{r=1}^{n} r^4=\frac{1}{30}n(n+1)(2n+1)(3n^2+3n-1)$. [6]

(b) Use the formula given in part (a) to find $50^4+51^4+\ldots+80^4$. [2]

13 In this question you must show detailed reasoning.

Use the formula $\sum_{r=1}^{n} r^2 = \frac{1}{6}n(n+1)(2n+1)$ to evaluate $121^2 + 122^2 + 123^2 + \ldots + 300^2$. [3]

14

(i) Show that $\frac{1}{2r-1}-\frac{1}{2r+5}\equiv\frac{6}{(2r-1)(2r+5)}$. [1]

Hence find

(ii) $\sum_{r=2}^{30}\frac{6}{(2r-1)(2r+5)}$, giving your answer correct to 3 decimal places, [5]

(iii) $\sum_{r=2}^{\infty}\frac{6}{(2r-1)(2r+5)}$, □□□giving your answer as a single fraction. [1]

15

Find $\sum_{r=1}^{n}(r^2 - r - 8)$, giving your answer in a fully factorised form. [5]

16

(a) Find $\sum_{r=1}^{n}\left(\frac{1}{r} - \frac{1}{r+2}\right)$. [3]

(b) What does the sum in part (a) tend to as $n \to \infty$? Justify your answer. [1]

17 Find $\sum_{r=1}^{n}(r+1)(r+5)$. Give your answer in a fully factorised form. [4]

18

(i) Show that $\sum_{r=n}^{2n} r^3 = \frac{3}{4}n^2(n+1)(5n+1)$. [4]

(ii) Hence find $\sum_{r=n}^{2n} r(r^2-2)$, giving your answer in a fully factorised form. [5]

19 Find $\sum_{r=1}^{n}(r-1)(r+1)$, giving your answer in a fully factorised form. [6]

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20

(i) Show that $\frac{1}{r} - \frac{3}{r+1} + \frac{2}{r+2} \equiv \frac{2-r}{r(r+1)(r+2)}$.

[2]

(ii) Hence show that $\sum_{r=1}^{n} \frac{2-r}{r(r+1)(r+2)} = \frac{n}{(n+1)(n+2)}$.

[5]

(iii) Find the value of $\sum_{r=2}^{\infty} \frac{2-r}{r(r+1)(r+2)}$.

[2]

21

The sequence $u_1, u_2, u_3, \ldots$ is defined by $u_1 = 2$ and $u_{n+1} = \frac{u_n}{1+u_n}$ for $n \geqslant 1$.

(i) Find u_2 and u_3, and show that $u_4 = \frac{2}{7}$.

[3]

(ii) Hence suggest an expression for u_n.

[2]

(iii) Use induction to prove that your answer to part (ii) is correct.

[5]

22

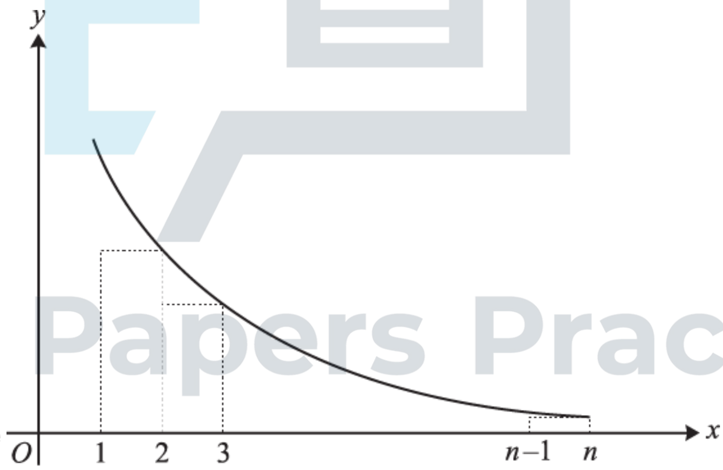
(i) By using a set of rectangles of unit width to approximate an area under the curve $y = \frac{1}{x}$, show that

$\sum_{x=1}^{\infty} \frac{1}{x}$ is infinite. [4]

(ii) By using a set of rectangles of unit width to approximate an area under the curve $y = \frac{1}{x^2}$, find an upper limit for the series $\sum_{x=1}^{\infty} \frac{1}{x^2}$. [5]

23

The diagram shows the curve $y = \frac{1}{x^3}$ for $1 \leqslant x \leqslant n$ where n is an integer. A set of $(n - 1)$ rectangles of unit width is drawn under the curve.



(i) Write down the sum of the areas of the rectangles.

[2]

(ii) Hence show that $\sum_{r=1}^{\infty} \frac{1}{r^3} < \frac{3}{2}$.

[5]

24

Find $\sum_{r=1}^{n}(3r+1)(r-1)$, giving your answer in a fully factorised form.

[5]

25

(i) Show that $\frac{1}{2r+1}-\frac{1}{2r+3}\equiv\frac{2}{(2r+1)(2r+3)}$.

[1]

(ii) Hence find $\sum_{r=1}^{n}\frac{1}{(2r+1)(2r+3)}$, giving your answer as a single fraction.

[6]

(iii) Find $\sum_{r=n}^{\infty}\frac{1}{(2r+1)(2r+3)}$, giving your answer as a single fraction.

[3]

26

Find $\sum_{r=1}^{n}(3r^2-5)$, expressing your answer in a fully factorised form.

[4]

27

(i) Show that $\frac{3}{r-1}-\frac{2}{r}-\frac{1}{r+1}\equiv\frac{4r+2}{r(r^2-1)}$. [2]

(ii) Hence find an expression, in terms of n, for $\sum_{r=2}^{n}\frac{4r+2}{r(r^2-1)}$.

[6]

(iii) Hence find the value of $\sum_{r=4}^{\infty}\frac{4r+2}{r(r^2-1)}$.

[2]

28

Find $\sum_{r=1}^{n}(4r^3 - 3r^2 + r)$, giving your answer in a fully factorised form.

[6]

29

(i) Show that $\frac{1}{3r-1} - \frac{1}{3r+2} \equiv \frac{3}{(3r-1)(3r+2)}$.

[2]

(ii) Hence show that $\sum_{r=1}^{2n} \frac{1}{(3r-1)(3r+2)} = \frac{n}{2(3n+1)}$.

[6]

30

(i) Show that $\frac{1}{r^2} - \frac{1}{(r+2)^2} \equiv \frac{4(r+1)}{r^2(r+2)^2}$.

[2]

(ii) Hence find an expression, in terms of n, for $\sum_{r=1}^{n} \frac{4(r+1)}{r^2(r+2)^2}$.

[6]

(iii) Find $\sum_{r=5}^{\infty} \frac{4(r+1)}{r^2(r+2)^2}$, giving your answer in the form $\frac{p}{q}$ where p and q are integers.

[2]

END OF QUESTION PAPER